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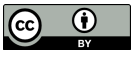
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Social Robots as Adaptive TEL Interfaces for 2e Learners: A Research Agenda

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Twice-exceptional (2e) learners are defined as individuals who are intellectually gifted while presenting one or more disabilities. This is a critically underserved population in Technology-Enhanced Learning (TEL) research. Their complex cognitive profiles can create design tensions that most current TEL systems fail to address, as standard adaptive platforms model learner ability along a single axis and offer little multimodal or socio-emotional support.

This position paper argues that social robots offer a distinctively promising interface paradigm for 2e learners, due to their physical presence, ability for multimodal interaction, and potential for real-time adaptive behavior. Drawing on sociocultural learning theory and recent advances in social robotics, we propose a research agenda organized around the three guiding questions of this workshop: the design and evaluation of inclusive TEL systems, the adaptation of such systems into real educational environments, and the multidimensional modeling of 2e user profiles. We discuss methodological patterns - including conversational analysis, participatory design, and longitudinal field studies - and call for cross-disciplinary collaboration to address this intersection of disability, giftedness and educational technology.

CCS Concepts: • **Human-centered computing** → **Field studies**; **Accessibility design and evaluation methods**; *Participatory design*; User models; • **Applied computing** → *Interactive learning environments*; • **Social and professional topics** → *People with disabilities*; Children; Adolescents.

Additional Key Words and Phrases: social robots, twice-exceptional learners, Technology-Enhanced Learning, inclusive HRI, inclusive HCI, adaptive interaction, special education, Human-Robot Interaction, Human-Computer Interaction

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1 Introduction

Twice-exceptional (2e) students occupy a paradoxical position in education, demonstrating outstanding ability in at least one domain while simultaneously meeting criteria for one or more disabilities [17]. Giftedness can mask disability, and disability can suppress the visible markers of giftedness, producing chronic misidentification and misallocation of support [11, 14]. The result is a population that fits poorly into standard educational categories, being too capable for special education interventions but too challenged for gifted programs, which consequently experiences chronic underachievement, disengagement, low self-esteem, and elevated rates of anxiety [2].

Technology-Enhanced Learning (TEL) systems have long promised personalized, adaptive instruction, yet their designs have not yet attended to 2e complexity. Most adaptive platforms model learning through a single ability dimension, typically error rate or knowledge state, which conflates the genuine competence of gifted students, with poor performance caused by an executive function deficit, a sensory overload event, or a processing speed gap [10].

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Screen-heavy, text-dependent interfaces compound, for instance, the problem for learners with dyslexia or attention regulation challenges, while offering too little intellectual challenge for their gifted dimension.

We argue that social robots, as multimodal agents capable of perceiving and adapting to social cues, offer a qualitatively different interface paradigm for this population. Unlike static chatbots, a social robot can serve as both scaffolding partner and intellectually engaging conversational peer, adapting its behavior in real time along multiple dimensions simultaneously. Crucially, to our knowledge, no empirical or design work has yet targeted 2e learners specifically in the context of social robotics, leaving a significant research gap at the intersection of inclusive education, HRI, HCI, and TEL.

2 Background

2.1 The 2e Profile: Strengths, Challenges, and the Design Gap

Defining 2e is itself contested. No consensus exists among researchers [18]: some characterize 2e students as those with *high abilities and co-existing learning difficulties* [7], while the most operationally used definition holds that 2e learners *demonstrate the potential for high achievement or creative productivity in one or more domains … AND manifest one or more disabilities* [17]. The difficulty is compounded by the contested nature of giftedness itself: a recent survey identified over 70 different criteria in the literature, with gifted students described simultaneously as *successful, confident, self-reliant high-performers* and as *loners who struggle, feel alienated, and are ridden by self-doubt* [21]. This definitional instability is not just theoretical: it has direct implications for how any TEL system might model and adapt to a 2e profile, and for what individuals are included in the group of 2e students.

Twice-exceptionality is oftentimes characterized by asynchronous development: advanced abstract reasoning or creative productivity might coexist with deficits in executive function, working memory, sensory regulation, or social communication [18]. Giftedness and disability interact synergistically rather than additively: the heightened meta-cognitive awareness typical of gifted learners can amplify the impact of executive-function failures, producing perfectionism, avoidance, and anxiety. Conversely, strong compensatory strategies delay identification of both the giftedness and the disability, sometimes for years [7].

Awareness of 2e learners is a recent development. Around the turn of the millennium, Brody and Mills [5] noted that many educators had difficulty "comprehending that a child can be gifted and also have learning disabilities"; similar misconceptions persist today [6, 14]. In Sweden, where our research will be conducted, the field is only beginning to emerge: gifted education research is expanding rapidly but starting from a very low baseline, and studies specifically addressing 2e students in Swedish schools remain rare [13]. This makes Sweden a timely and practically significant context for this work. Designing TEL systems for this population is accordingly a challenge with very little prior art to draw on: the requirement is not to support disability *or* giftedness, but both simultaneously and in ways sensitive to how each dimension shapes the other.

2.2 Social Robots in Education: Evidence and Promise

A now-substantial body of research documents the effectiveness of social robots as educational agents. Belpaeme et al.'s landmark review [4] established that robots can match human tutoring on restricted tasks and that physical embodiment produces distinct advantages: users display more social behaviors (joint attention, turn-taking, reciprocal disclosure) with a physically present robot than with a virtual agent, and learning gains are consistently higher [15].

A recent meta-analysis of 146 studies [23] found a strong overall positive effect against no-training control conditions (Cohen's $d = 0.75$) and identified the co-teacher configuration (robot alongside a human educator rather than as a replacement) as particularly effective ($d = 0.88$).

For language and communication learning specifically, the evidence is especially encouraging. Meta-analyses of robot-assisted language learning [8, 23] consistently report positive effects, particularly for vocabulary in children. The theoretical grounding is Vygotsky's [22] account of learning as fundamentally interactional: a robot capable of sustained, linguistically calibrated dialogue can serve as a *More Knowledgeable Other* (MKO), scaffolding learners through their Zone of Proximal Development by adjusting challenge, offering feedback, and maintaining motivating engagement.

Socially assistive robots have also been studied extensively with neurodiverse populations. For children on the autism spectrum, NAO and Pepper robots have demonstrated consistent improvements in joint attention, emotion recognition, conversational turn-taking, and social engagement [19]. The predictability and consistency of a robot appears to reduce social anxiety and lower the threshold for interaction [19]. A smaller literature addresses ADHD, where robots show promise for attention regulation and task persistence [4, 16]. Despite this breadth of work with disabled people, *no study has specifically targeted twice-exceptional learners with social robots*, whose design requirements might differ qualitatively from those of either gifted-only or disability-only populations [18].

3 Research Agenda

We organize our proposed research agenda around the three guiding questions of the workshop. Together they outline a PhD project that begins with theoretical synthesis, moves through participatory design, and culminates in longitudinal field studies.

3.1 Q1: Designing and Evaluating Inclusive Social-Robot TEL Systems

How can we design and evaluate TEL systems that promote effective learning, well-being, and inclusivity with innovative HCI?

Standard evaluation metrics for educational technology - task completion, knowledge tests, time on task - are insufficient for 2e learners. A student may score well on a knowledge test despite experiencing overwhelming stress or chronic disengagement; on the other hand, a student may appear off-task while engaged in the deep associative thinking characteristic of giftedness. We propose an evaluation framework integrating three complementary dimensions: *intellectual engagement* as depth and originality of responses, and initiative in questioning; *regulatory efficacy* as on-task persistence, self-reported stress, and detected frustration episodes); and *long-term well-being* as academic self-concept, school belonging, and reduced avoidance behaviors. Grounding this framework in an inclusive perspective requires conceptualizing inclusion not merely as placement but as active participation [9].

The interaction model we term *dual-adaptive interaction* responds simultaneously to both dimensions of the 2e profile. On the *cognitive axis*, the robotic platform dynamically modulates task difficulty, introduces creative extensions, and poses Socratic questions calibrated to demonstrated strengths [1, 12]. On the *regulatory axis*, it monitors behavioral indicators of frustration, sensory overload, or distraction - through affect recognition, gaze tracking, and speech analysis - and responds with timely interventions: structured micro-breaks, varied interaction modalities, or motivational reframing. The key design challenge is managing the interplay between these axes: increasing cognitive challenge may elevate regulatory demand, while excessive scaffolding risks signaling low expectations to a learner whose gifted dimension depends on intellectual stimulation and autonomy.

Participatory design with 2e students, their caregivers, and teachers is central to our approach. This community has been underrepresented in HCI design processes; involving them in scenario development and prototype feedback will both produce more ecologically valid design choices and align with the ethical imperative of designing *with* rather than *for* marginalized populations [3].

3.2 Q2: Adapting to the Complexity of Real Educational Environments

How can we adapt TEL systems to the complexity of educational environments?

Laboratory evaluations of social robots rarely capture the full complexity of the real learning environment (classroom): noise, competing social dynamics, unpredictable interruptions, shifting group configurations, and the presence of human teachers whose roles must be carefully coordinated with those of the robot. For 2e learners this complexity might be compounded: the same student may need one-on-one regulatory support in one session and intellectually challenging group facilitation in another.

Our research agenda foregrounds ecological validity. Initial studies will take place in authentic school settings in Sweden, a national context where 2e awareness is growing but educational research and practice remain nascent [13]. Studies will examine role allocation between the robot and the classroom teacher, with attention to how human-robot collaboration can preserve teacher authority and professional judgment while leveraging the robot's capacity for consistent, patient, individualized interaction. A cross-national comparison between Swedish and Italian educational contexts - two systems with markedly different approaches to giftedness identification and special needs provision - will further help understanding how institutional environment shapes what *complexity of the educational environment* means for this population in practice.

We will apply Conversation Analysis (CA) as a primary methodological tool [20]. CA's focus on the moment-by-moment organization of talk-and-gesture sequences makes it uniquely suited to examining *how* interaction unfolds in child-robot-teacher triads: who initiates, how scaffolding is delivered and received, where breakdowns occur, and how they are repaired. This fine-grained lens complements broader experimental designs and provides an accountability structure for the robot's adaptive behavior.

3.3 Q3: Modeling the 2e Learner in TEL Systems

How can we model users to account for their diversity, preferences, and constraints?

Standard learner models collapse ability onto a single proficiency dimension, making them structurally incapable of representing a student who excels at abstract conceptualization but struggles with sustained attention or written output. We propose a *multidimensional 2e learner model* that maintains interacting sub-models for: (a) domain knowledge and cognitive ability, (b) executive function state (attention, task initiation, working memory load), (c) sensory and affective state (frustration, sensory overload, boredom), and (d) communicative preferences (preferred modalities, pace, degree of social engagement).

These sub-models must update in real time from multimodal sensing while handling the inherent ambiguity of 2e data: a student's observable behavior (appearing inattentive, producing brief responses) is often a poor proxy for underlying ability. Robust uncertainty handling, possibly through Bayesian or neural approaches, is essential to avoid the dual risk of over-scaffolding (which suppresses gifted engagement) and under-scaffolding (which leaves the disability dimension unsupported).

Ethical considerations are foundational here. Collecting and processing disability-related behavioral data on minors requires explicit informed consent, stringent privacy protections, and transparency in how the model is used.

We are particularly attentive to the risk of algorithmic stereotyping: a model that conflates low processing speed with low ability, or social withdrawal with disinterest, would reproduce in computational form the same misidentification that 2e students currently experience from human educators.

4 Conclusion

Twice-exceptional learners challenge existing frameworks in both education and educational technology. They require an interface that is simultaneously enriching and supportive, consistent and adaptive, patient and intellectually stimulating. We have argued that social robots constitute a uniquely promising design space for meeting this dual mandate.

The research agenda outlined here responds to the three guiding questions of the EduIHM workshop and is grounded in sociocultural theory, evidence-based robotics research, and a commitment to participatory, ecologically valid methods. We hope this paper will stimulate dialogue with the HCI-for-TEL community and initiate collaborations that can accelerate progress at this neglected intersection of giftedness, disability, and educational technology.

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